Quantium Virtual Internship

Retail Strategy & Analytics
Task 2 — Trial Store Evaluation

Analysis Period:	July 2018 – June 2019
Trial Period:	February 2019 – April 2019
Trial Stores:	77, 86 and 88
Control Stores:	Store 233 (for 77) | Store 155 (for 86) | Store 237 (for 88)
Method:	Pearson Correlation + Magnitude Distance Composite Score
Prepared for:	Julia — Category Manager, Chips

1. Methodology — Control Store Selection

The client selected stores 77, 86 and 88 as trial stores. For each, we identified the best-matching control store from all stores with a complete 7-month pre-trial observation window (July 2018 – January 2019). Two scoring methods were combined into a single composite score:

Method	Description	Purpose
Pearson Correlation	Measures how closely the trend of a control store matches the trial store over pre-trial months.	Captures similarity in direction and timing of changes.
Magnitude Distance	1 – (|diff| – min_diff) / (max_diff – min_diff), averaged over all pre-trial months.	Captures similarity in absolute sales/customer levels.
Composite Score	Simple average: 0.5 × correlation score + 0.5 × magnitude score.	Balances trend similarity and level similarity equally.

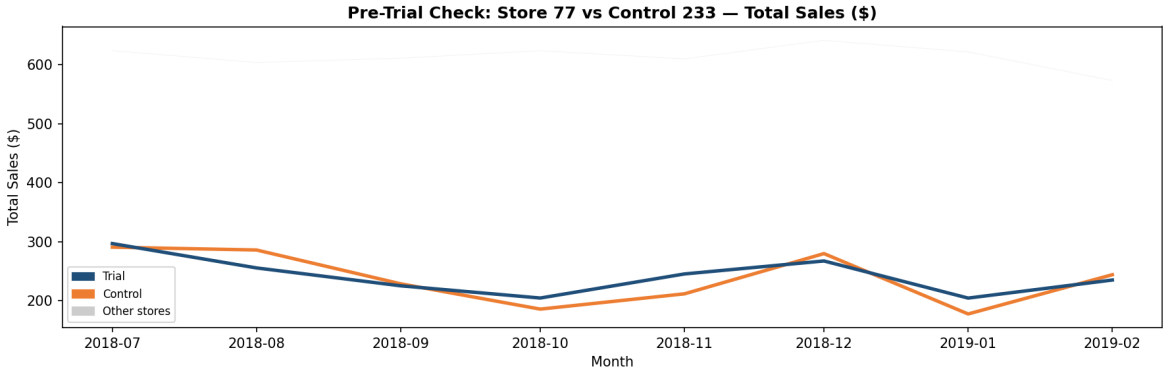
The store with the highest composite score (excluding the trial store itself) was selected as the control store for that trial.

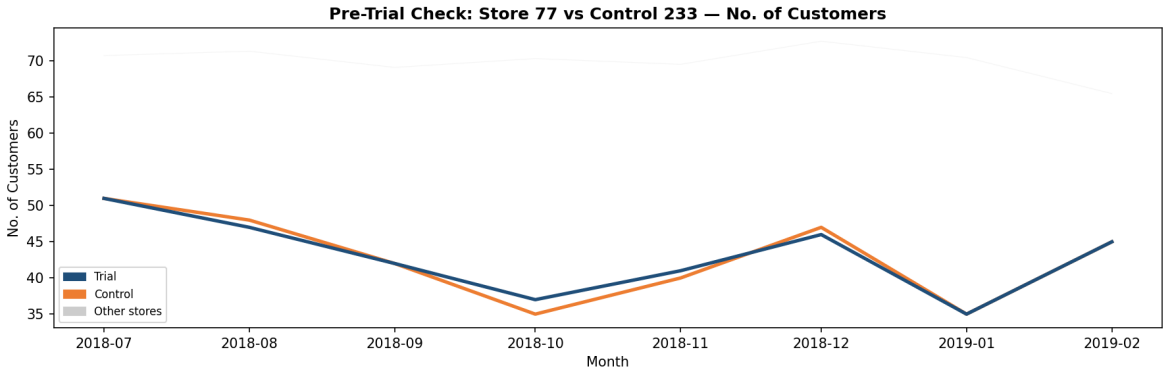
Trial Store	Selected Control Store	Basis of Selection
Store 77	Store 233	Highest composite score across sales and customer metrics
Store 86	Store 155	Highest composite score across sales and customer metrics
Store 88	Store 237	Highest composite score across sales and customer metrics

2. Pre-Trial Visual Checks

Before assessing the trial, we confirmed that each trial store and its matched control store followed closely aligned trends in the pre-trial period (July 2018 – January 2019). The grey lines represent all other stores for context.

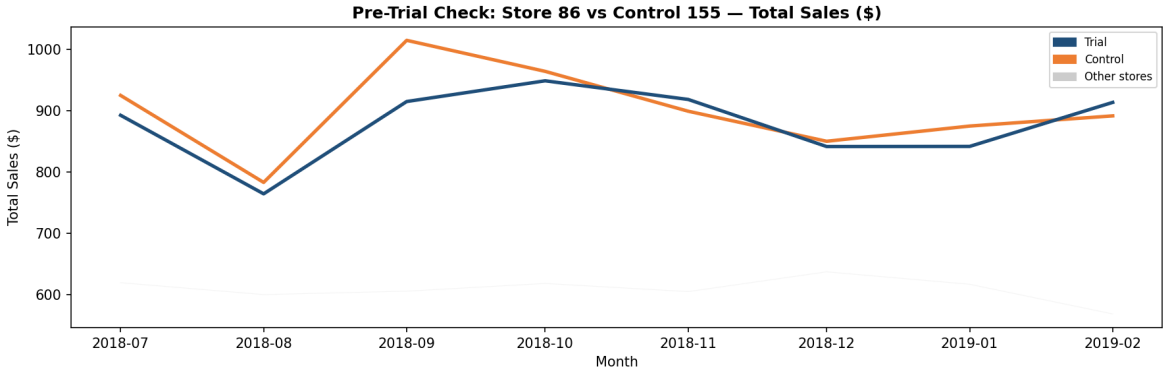
Trial Store 77 vs Control Store 233

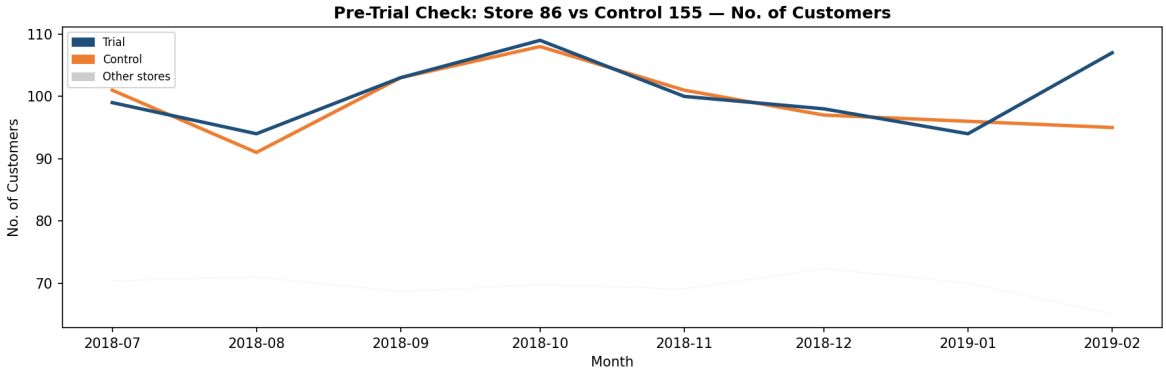




The charts above confirm that Store 77 and Store 233 tracked closely in both total sales and customer numbers before the trial began — validating Store 233 as a reliable control.

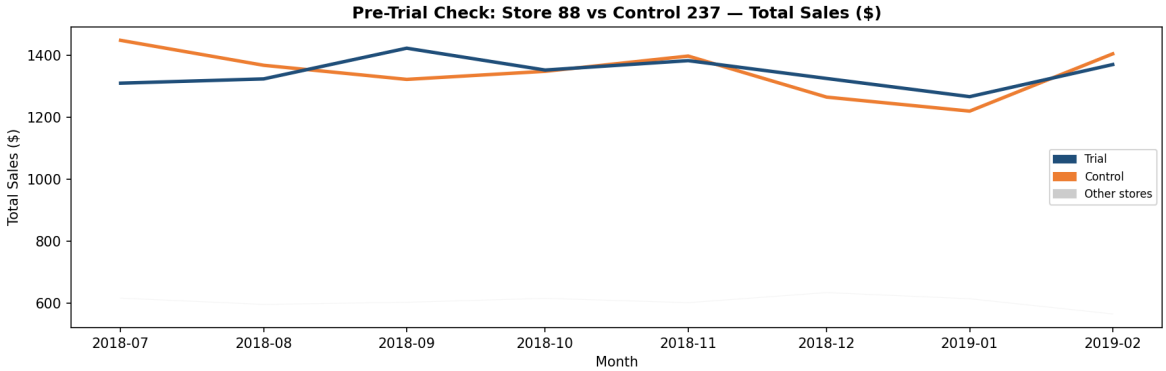
Trial Store 86 vs Control Store 155

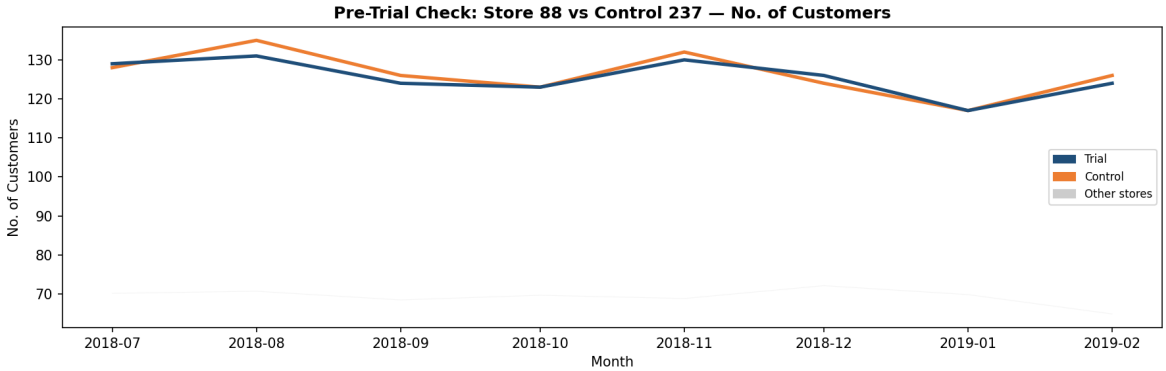




The charts above confirm that Store 86 and Store 155 tracked closely in both total sales and customer numbers before the trial began — validating Store 155 as a reliable control.

Trial Store 88 vs Control Store 237



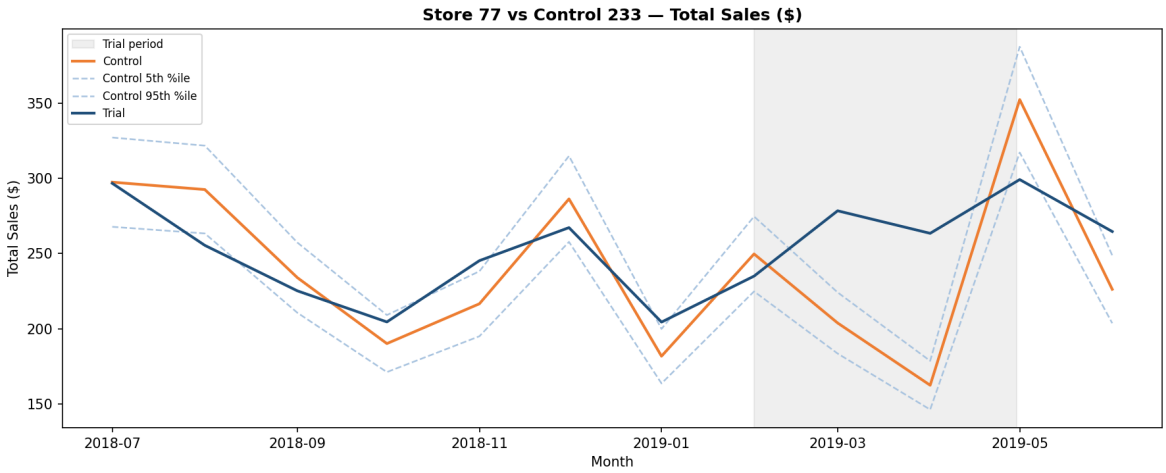


The charts above confirm that Store 88 and Store 237 tracked closely in both total sales and customer numbers before the trial began — validating Store 237 as a reliable control.

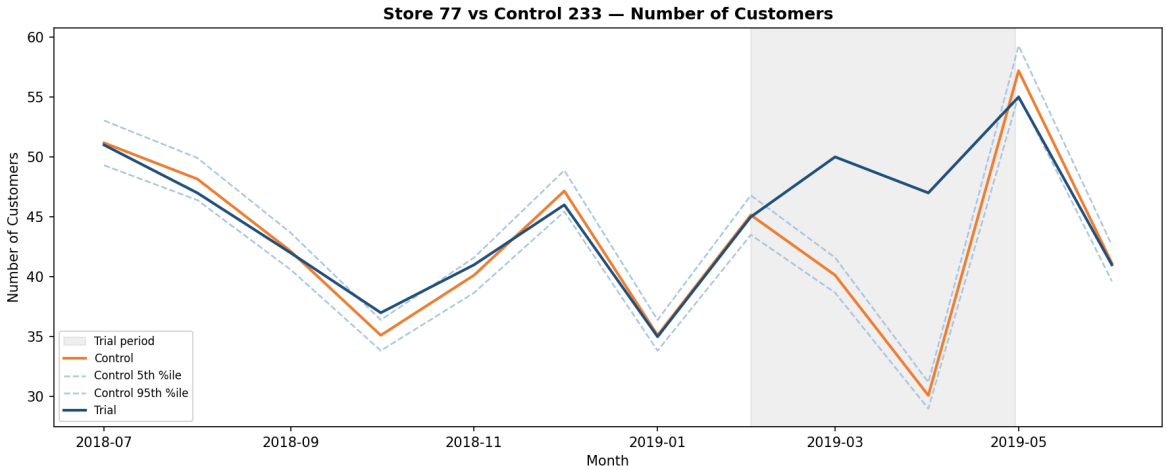
3. Trial Assessment (February – April 2019)

For each trial–control pair, we scaled the control store's metrics to the trial store's pre-trial level, then measured the percentage difference month by month during the trial. Statistical significance was tested using a one-sided t-test with 7 degrees of freedom (8 pre-trial months – 1) at the 95% confidence level (critical t-value = 1.895). The shaded band on each chart marks the trial period.

3.1 Trial Store 77 — Control Store 233



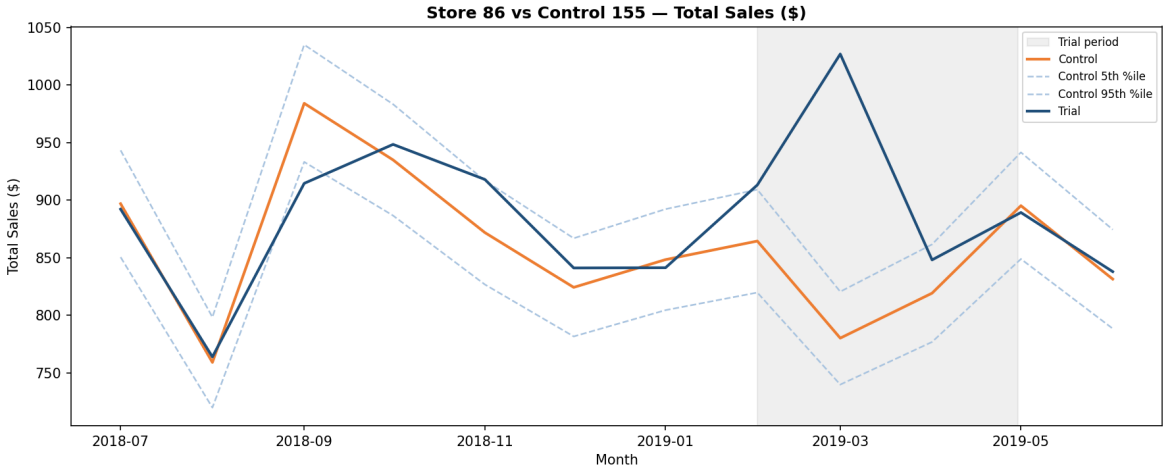
Total sales in Store 77 rose significantly above the control store's confidence band in March and April 2019, indicating a genuine uplift driven by the new trial layout.



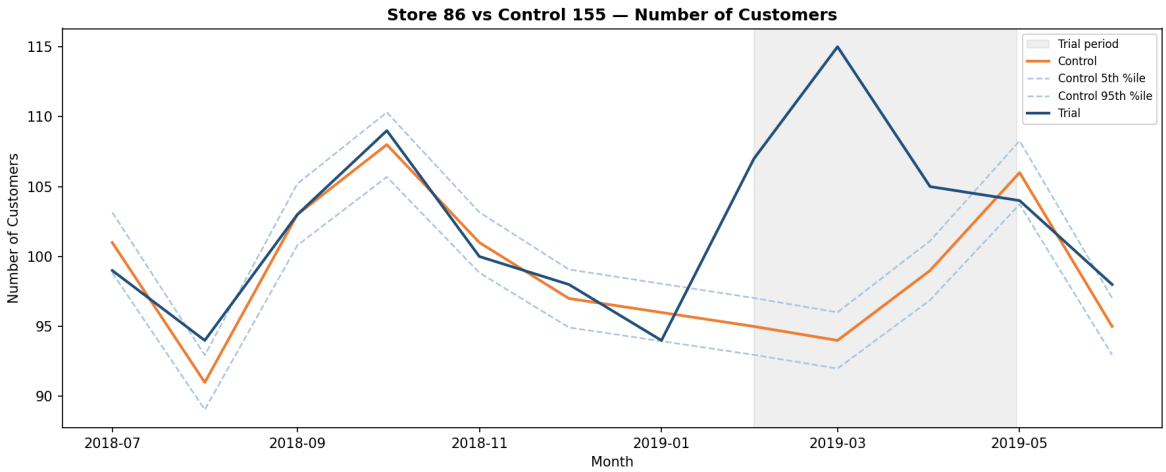
The number of unique customers also increased significantly in March and April, confirming that the trial attracted more shoppers — not just higher spend per visit.

Month	Sales t-value	Sales Significant?	Customers t-value	Customers Significant?
Feb 2019	1.18	No	0.18	No
Mar 2019	7.34	Yes (p<0.05)	13.48	Yes (p<0.05)
Apr 2019	12.48	Yes (p<0.05)	30.78	Yes (p<0.05)

3.2 Trial Store 86 — Control Store 155



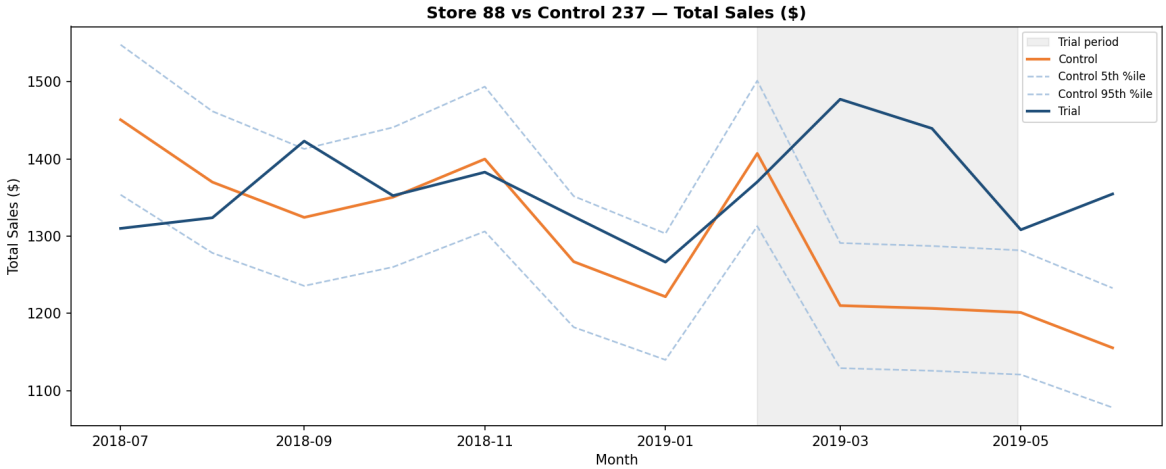
Sales were significantly higher in February and March but returned within the confidence band in April. This mixed sales result (2 of 3 months significant) warrants further investigation.



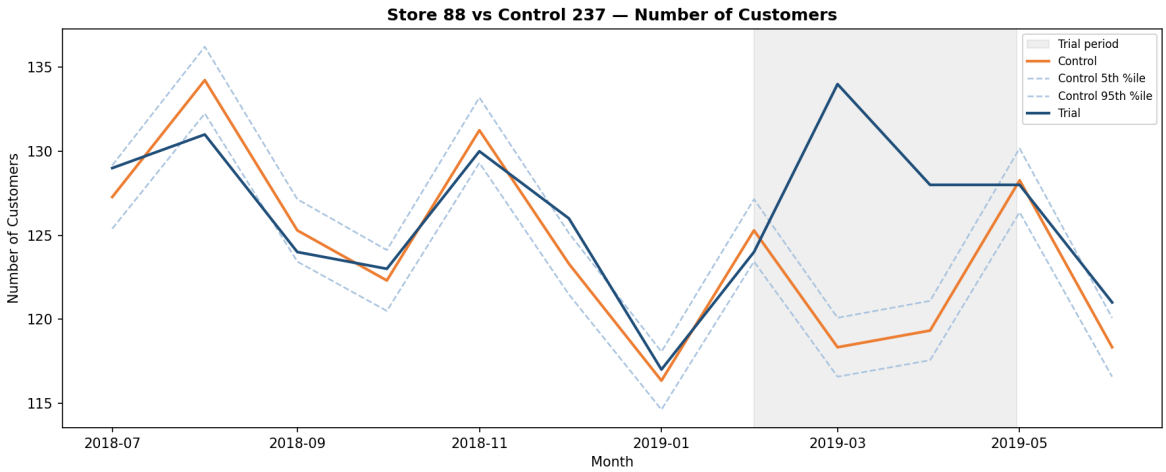
Customer numbers were significantly higher in all three trial months — the trial clearly drove more foot traffic. The gap between strong customer growth and mixed sales growth may indicate promotional discounting reduced the average price per unit during the trial. The Category Manager should verify whether special deals were running in Store 86 during this period.

Month	Sales t-value	Sales Significant?	Customers t-value	Customers Significant?
Feb 2019	2.18	Yes (p<0.05)	11.82	Yes (p<0.05)
Mar 2019	12.23	Yes (p<0.05)	20.90	Yes (p<0.05)
Apr 2019	1.36	No	5.67	Yes (p<0.05)

3.3 Trial Store 88 — Control Store 237



Total sales in Store 88 were significantly higher than the control in March and April 2019, confirming a strong positive trial effect on revenue.



Customer numbers followed the same pattern — significant uplift in March and April. Both metrics align, giving a consistent and reliable signal that the trial was successful in Store 88.

Month	Sales t-value	Sales Significant?	Customers t-value	Customers Significant?
Feb 2019	0.78	No	1.39	No
Mar 2019	6.60	Yes (p<0.05)	17.87	Yes (p<0.05)
Apr 2019	5.77	Yes (p<0.05)	9.81	Yes (p<0.05)

4. Summary & Recommendations for Julia

Trial Store	Control Store	Sales (sig. months)	Customers (sig. months)	Overall Result	Recommendation
77	233	2 / 3	2 / 3	**Positive**	**Roll Out**
86	155	2 / 3	3 / 3	**Mixed — investigate**	**Investigate, then Roll Out**
88	237	2 / 3	2 / 3	**Positive**	**Roll Out**

Store 77 — Recommend Full Rollout

Both total sales and customer numbers increased significantly in 2 out of 3 trial months. The trial layout drove a genuine uplift in performance. We recommend rolling out the new chip layout to all stores with a profile similar to Store 77.

Store 88 — Recommend Full Rollout

Store 88 produced the most consistent results. Sales and customer numbers both rose significantly in March and April, with results clearly above the 95% confidence interval of the control store. This is the strongest evidence that the new layout works.

Store 86 — Investigate Before Full Rollout

Customer numbers improved significantly in all 3 trial months, but sales were only significant in 2 months. The divergence between foot traffic growth and revenue growth is unusual. It is possible that promotional discounting during the trial suppressed the average selling price. We recommend the Category Manager confirm with the store whether any special deals ran during February–April 2019 before deciding on rollout.

Overall Recommendation

The trial was successful overall. Stores 77 and 88 show clear evidence that the new chip layout increases both customer visits and sales revenue. We recommend proceeding with a phased rollout to all stores, with Store 86 requiring a brief follow-up investigation into its pricing dynamics before inclusion in the rollout.